

On the Stability of Internal Baroclinic Jets in a Rotating Atmosphere¹

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ABSTRACT

We consider the quasi-geostrophic instability of a circumpolar vortex in which there is available kinetic energy of lateral shear as well as available potential energy due to meridional temperature gradients. Stability criteria are developed for the case of an internal jet, i.e., where the meridional temperature gradients at the ground vanish. The internal jet is stable if the gradient of potential vorticity in isentropic surfaces does not vanish. If it vanishes at a closed isopleth of constant mean zonal vorticity, the jet is unstable.

The special role played by the kinematic and thermodynamic boundary conditions in the theory of baroclinic stability is clarified by re-examining earlier theories in the light of an analogy to two-dimensional shear flow. Simple baroclinic flow with rigid horizontal boundaries is isomorphic to Couette flow with free boundaries. The presence of meridional temperature gradients at boundaries relaxes the constraints on the boundary pressure perturbations and makes possible the release of available potential energy, just as the freeing of the boundary in Couette flow makes possible the release of shear kinetic energy.

The mid-winter breakdown of the polar-night jet may be an example of an instability, but we cannot say whether the concomitant disturbance releases mean kinetic, mean potential energy, or both, of the internal jet type, since the possibility of all of these conversions exists. The apparent downward propagation of the unstable disturbances in the polar-night jet below 30 km may be explained by the prior onset of instability at higher levels.

1. Introduction and summary

Axisymmetric heating of the atmosphere of a rotating spherical planet will tend to produce an axisymmetric vortex whose flow will be predominantly zonal if the rotation rate is sufficiently large. The mean circulation of the earth's atmosphere approximates to this condition, and the formation of cyclonic disturbances can be attributed to the instability of the zonal flow to axially asymmetric perturbations. A considerable literature on the stability of zonal flow has arisen in the past 40 years or so, inspired in part by the early stability investigations of Helmholtz and Rayleigh. In recent years attention has turned from frontal discontinuities to continuous horizontal temperature gradients as representing more realistic sources of energy for the disturbances containing the bulk of the atmosphere's energy. The investigations have been

confined to flows in which the horizontal temperature gradients are uniform, and in which, consequently, there is vertical but no horizontal shear in the zonal flow. This idealization of the stability problem for a zonal vortex is known as the problem of baroclinic stability, and has been treated by Charney (1947), Eady (1949), Fjörtoft (1950), Kuo (1952), Green (1960), Burger (1962) and others. The source of the energy for the growing disturbances is the potential energy associated with horizontal temperature gradients rather than the kinetic energy of the shearing motion; horizontal shear is absent, and the vertical Reynolds stresses in motions of planetary scale are too small to convert the kinetic energy of vertical shear.

In reality, horizontal temperature gradients are never uniform and the balance of forces in a stationary zonal vortex requires horizontal as well as vertical shear. There may be instances in which the horizontal shear plays a decisive dynamical role. One such instance is presented by the so-called "polar-night jet." This is a circumpolar vortex which forms in the fall and reaches

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its maximum intensity in the polar winter when the absence of heating at very high latitudes, combined with intense ozone absorption of sunlight at lower latitudes and lack of meridional heat exchange, produces strong stratospheric temperature gradients near latitude 60° and a wind maximum at about 60 km. There is no corresponding high-latitude jet in the troposphere, where the absence of ozone absorption and the strong meridional heat exchange brought about by cyclonic activity considerably reduce the meridional temperature gradients.

The polar-night jet develops minor perturbations in the fall, but in late winter or early spring it undergoes a rapid warming and simultaneously breaks down into a number of smaller circulations. The suddenness and rapidity of the transition suggests some form of hydrodynamic instability. Murray (1960) investigated the baroclinic stability of an idealized atmosphere consisting of a troposphere without horizontal temperature gradients and a stratosphere with a uniform horizontal temperature gradient. He found, somewhat surprisingly, that the flow is stable despite the large store of potential energy available for conversion into kinetic energy. Murray then examined the stability of the polar-night jet for purely horizontal displacements at the 25-mb level (about 25 km) and found the flow there to satisfy the Rayleigh necessary condition for instability, viz., that the meridional gradient of the vertical component of absolute vorticity must vanish at some point. This criterion has been shown by Kuo (1949) and Fjörtoft (1950) to be the analogue for the rotating sphere of Rayleigh's criterion (1880) for plane parallel flow in an inviscid fluid. Although purely horizontal displacements are not possible in a baroclinic atmosphere, Murray's finding emphasizes the dynamical importance of horizontal shear.

The analysis of the stability of zonal flow with both horizontal and vertical shear is complicated by the non-separability of the perturbation eigenfunction equation, reflecting a coupling of two physically different energy releasing mechanisms. However, in the special case of an "internal jet," where the potential temperature is constant at rigid horizontal boundaries, we have been able to deduce a necessary criterion for instability by generalizing the integral technique of Rayleigh (*loc. cit.*). The generalization states that a zonal internal jet is stable to quasi-geostrophic wave perturbations if the horizontal variation of potential vorticity, measured in isentropic surfaces, never vanishes. In Murray's model the latter quantity is everywhere positive, and it can be shown that the presence of a shearing discontinuity at the tropopause does not alter the validity of this criterion. We may therefore conclude immediately that his flow is stable. If the polar-night jet is regarded as an internal jet, and the horizontal shear is taken into account, it is found that the meridional potential vorticity gradient

does develop sign changes as the winter progresses. This suggests that the breakdown may be due to an instability, although not a purely baroclinic one.

The simplest model to which our criterion applies is a geostrophically balanced free jet in an incompressible homogeneous layer of fluid resting on an infinitely deep layer of heavier fluid (Stern, 1962). If the potential vorticity, defined as the ratio of the absolute vorticity to the depth of the upper layer, varies monotonically across the stream, there can be no quasi-geostrophic instability.

Apparently the existence of a meridional temperature gradient at the earth's surface is a distinct destabilizing factor, so that its absence may prevent or inhibit the release of available potential energy in the free atmosphere. Indeed one encounters the paradoxical result that a baroclinic flow with a uniform horizontal potential temperature gradient is stable in the absence of constraining horizontal boundaries but becomes unstable when horizontal boundaries are introduced. In seeking to resolve this paradox we have discovered a close parallelism between the stability properties of zonal flow in a baroclinic fluid with uniform horizontal potential temperature gradient and the stability properties of plane parallel flow in an incompressible inviscid fluid. By means of this parallelism the theory of baroclinic stability is reduced to the conceptually simpler and more extensively investigated theory for plane parallel flow. The physical explanation of the destabilizing effect of boundaries is then readily found. The introduction of a rigid boundary in the atmospheric case is analogous to the replacement of the strong constraint of vanishing energy density at infinity, or of vanishing stream-function at a rigid boundary, by the weaker one of vanishing pressure at a free surface. The former require the Reynold's stress to vanish, whereas the latter does not, and we can show, by generalizing a result of Foote and Lin, that the vanishing of the Reynold's stress at boundaries forces it under certain circumstances to vanish everywhere. The condition of vanishing temperature gradient at a rigid boundary in the atmosphere is analogous to the rigid boundary condition in plane flow. Permitting the temperature gradient to differ from zero relaxes the boundary constraint and allows the atmospheric quantity which is analogous to the Reynold's stress (horizontal eddy flux of heat) to be different from zero.

2. Derivation of the quasi-geostrophic perturbation equations

Let L be a characteristic horizontal dimension, D a characteristic vertical dimension, U a characteristic particle velocity relative to the rotating earth and Ω the angular speed of the earth's rotation. We shall be concerned with motions for which D/L and the Rossby number $Ro = U/2\Omega L$ are so small that the departures from hydrostatic and geostrophic equilibrium are

small. The equations governing such motions have been derived in a more or less general form, although not in an entirely consistent manner, by Charney (1948) and Eliassen (1949), independently. A more consistent derivation has been given by Burger (1958) and Charney (1960). We shall briefly recapitulate Charney's derivation in the form most suitable for our present purposes.

Let p be the pressure, ρ the density, α the specific volume, g the acceleration of gravity (assumed constant), a the radius of the earth, ϕ the latitude, $\mathbf{V}$ the relative horizontal particle velocity component, w the vertical velocity component, ∇ the horizontal component of the gradient operator, z the upward directed vertical coordinate and $\mathbf{k}$ a unit vertical vector. Dimensional *variables* (as distinguished from constants) will be denoted by an asterisk superscript, and horizontal space-time averages will be denoted by the subscript "s." It is assumed that p_s^* and ρ_s^* are hydrostatically related and that $\alpha_s^* = 1/\rho_s^*$.

For a perfect gas the specific entropy is

$$c_p \ln \theta^* = c_p \ln p^* + c_p \ln \alpha^* + \text{constant}, \quad (2.1)$$

where θ^* is the potential temperature (constant for an adiabatic process), and c_p and c_v are the specific heats at constant pressure and constant volume, respectively. The quantity $g \partial \ln \theta_s^* / \partial z$ is the square of the mean frequency of a vertical buoyancy oscillation and measures the gravitational stability of the fluid. We assume that it is everywhere positive.

The following non-dimensional parameters are found to be relevant:

$$\begin{aligned} Ro &= \frac{U}{2\Omega L}, & \delta &= \frac{D}{L}, & \mu &= \frac{D}{H^*}, \\ \kappa &= \frac{D \partial \ln \theta_s^*}{\partial z^*}, & \epsilon &= \frac{4\Omega^2 L^2}{g D \kappa}, & \gamma &= \frac{c_p}{c_v}, \end{aligned} \quad (2.2)$$

where H^* is the scale-height,

$$H^* = p_s^* \alpha_s^* / g. \quad (2.3)$$

The last two non-dimensional parameters could be replaced by the Froude and Richardson numbers,

$$\left. \begin{aligned} F &= \frac{U^2}{g D} = \epsilon \kappa Ro^2, \\ Ri &= \frac{g D \kappa}{U^2} = \epsilon^{-1} Ro^{-2}. \end{aligned} \right\} \quad (2.4)$$

The range of the non-dimensional parameters will be limited by explicit assumption so as to focus attention on certain species of instabilities which are of interest

from the point of view of the large scale dynamics. We are interested in modes which

- $$\left. \begin{aligned} (a) & \text{ are quasi-hydrostatic } (\delta \ll 1) \\ (b) & \text{ are quasi-geostrophic } (Ro \ll 1, \text{ and the characteristic time scale is not greater in order of magnitude than the advective time scale } L/U) \\ (c) & \text{ are potential energy converting } (\epsilon \sim 1 \text{ or greater}) \\ (d) & \text{ are quasi-Boussinesq } (\kappa \ll 1, \kappa \epsilon \ll 1, \mu/\gamma \sim 1 \text{ or less}) \\ (e) & \text{ satisfy the beta plane approximation } (L/a \sim Ro \text{ or less}). \end{aligned} \right\} \quad (2.5)$$

The typical magnitudes, $Ro = 10^{-1}$, $L/a = 10^{-1}$, $\delta = 10^{-2}$, $\mu/\gamma = 1$, $\kappa = 10^{-1}$, $\epsilon = 1$, indicate that the above assumptions are satisfied with more or less accuracy in the actual atmosphere.

Assumption (c) in combination with (a) and (b) allows the possibility of a class of waves for which the order of magnitude of the slope of the particle trajectories is less than or equal to that of the isentropic surfaces. Such motions can convert potential energy into kinetic energy, or vice versa. Thus, in the baroclinic model studied by Eady (1949), where the only possible source of perturbation energy is the potential energy of the mean flow, the criterion for instability can be expressed as $\epsilon \sim 1$, or $L \sim L_e$, where L_e is the Rossby "radius of deformation," $[(4\Omega^2)^{-1} g \kappa D]^{\frac{1}{2}}$. Motions for which the gravitational stability is so large that $\epsilon \ll 1$ will not be considered. To the first order of approximation they are merely horizontal incompressible flows (Charney, 1960).

It will be convenient to express the horizontal velocity $\mathbf{V}^*$ as the sum of a non-divergent part $\mathbf{V}_\psi^* = \mathbf{k} \times \nabla^* \psi^*$ and an irrotational part $\mathbf{V}_\sigma^* = \nabla^* \sigma^*$.

On the basis of the foregoing definitions and assumptions it is possible to define non-dimensional forms of the dynamical variables in such a way that they have orders of magnitude unity. The required definitions are

$$\left. \begin{aligned} \nabla^* &\equiv L^{-1} \nabla, & z^* &\equiv D z, & t^* &\equiv L U^{-1} t, \\ \mathbf{V}^* &\equiv U \mathbf{V}, & \mathbf{V}_\psi^* &\equiv U \mathbf{V}_\psi, & \psi^* &\equiv L U \psi, \\ \mathbf{V}_\sigma^* &\equiv Ro U \mathbf{V}_\sigma, & \sigma^* &\equiv Ro L U \sigma, & w^* &\equiv \delta Ro U w, \\ p^* &\equiv p_s^* (1 + \mu \kappa \epsilon Ro p), \\ \rho^* &\equiv \rho_s^* (1 + \kappa \epsilon Ro p), \\ \alpha^* &\equiv \alpha_s^* (1 + \kappa \epsilon Ro \alpha), \\ \ln \theta^* &\equiv \ln \theta_s^* + \kappa \epsilon Ro \theta. \end{aligned} \right\} \quad (2.6)$$

The dimensional forms of the equations of relative motion for an inviscid fluid moving under the influence of gravity are

$$\left(\frac{\partial}{\partial t^*} + \mathbf{V}^* \cdot \nabla^* + w^* \frac{\partial}{\partial z^*} \right) \mathbf{V}^* + 2\Omega \sin \phi \mathbf{k} \times \mathbf{V}^* + 2\Omega \cos \phi w^* \mathbf{i} = -\alpha^* \nabla^* p^*, \quad (2.7)$$

$$\left(\frac{\partial}{\partial t^*} + \mathbf{V}^* \cdot \nabla^* + w^* \frac{\partial}{\partial z^*}\right) w^* - 2\Omega \cos\phi \mathbf{V}^* \cdot \mathbf{i} = -\alpha^* \frac{\partial p^*}{\partial z^*} - g, \quad (2.8)$$

where $\mathbf{i}$ is a unit vector directed from west to east. Substituting from (2.6) and neglecting small quantities, one obtains immediately the "zero-order" (non-dimensional) geostrophic and hydrostatic approximations:

$$\sin\phi \mathbf{k} \times \mathbf{V} \approx -\nabla p, \quad (2.9)$$

$$\frac{\alpha}{\alpha_s^*} \approx -\frac{\rho}{\alpha_s^*} \approx -\frac{\partial}{\partial z} \left(\frac{p}{\alpha_s^*} \right). \quad (2.10)$$

These equations are diagnostic rather than prognostic because of the absence of time derivatives. The non-dimensional time dependent equations are obtained by taking the horizontal divergence and curl of Eq (2.7) and substituting from (2.6). It is convenient first to discard the third term on the left-hand side of (2.7) as being small of order δ compared with the first and small of order δRo compared with the second. One then obtains the divergence equation,

$$(1 + \kappa \epsilon Ro \alpha) \nabla^2 p + \kappa \epsilon Ro \nabla \alpha \cdot \nabla p - \nabla \cdot (\sin\phi \nabla \psi) + Ro \nabla \cdot [\mathbf{V}_\psi \cdot \nabla \mathbf{V}_\psi + \mathbf{k} \times \mathbf{V}_\sigma \sin\phi] + Ro^2 \nabla \cdot [\partial \mathbf{V}_\sigma / \partial t + \mathbf{V}_\psi \cdot \nabla \mathbf{V}_\sigma + \mathbf{V}_\sigma \cdot \nabla \mathbf{V}_\psi + w \partial \mathbf{V}_\psi / \partial z] + Ro^2 \nabla \cdot [\mathbf{V}_\sigma \cdot \nabla \mathbf{V}_\sigma + w \partial \mathbf{V}_\sigma / \partial z] = 0, \quad (2.11)$$

and the vorticity equation,

$$(\partial / \partial t + \mathbf{V}_\psi \cdot \nabla) \nabla^2 \psi + Ro^{-1} \mathbf{V}_\psi \cdot \nabla \sin\phi + \nabla \cdot (\mathbf{V}_\sigma \sin\phi) + \kappa \epsilon (\nabla \alpha \times \nabla p \cdot \mathbf{k}) + Ro \nabla \cdot (\mathbf{V}_\sigma \nabla^2 \psi + w \partial \nabla \psi / \partial z) + Ro^2 \nabla \cdot [w \partial (\mathbf{V}_\sigma \times \mathbf{k}) / \partial z] = 0. \quad (2.12)$$

If the dependent variables p , α , ψ , σ and w are expanded as a power series in Ro the first approximation equations become

$$\nabla \cdot (\sin\phi \nabla \psi) = \nabla^2 p, \quad (2.13)$$

and

$$(\partial / \partial t + \mathbf{V}_\psi \cdot \nabla) \nabla^2 \psi + Ro^{-1} \mathbf{V}_\psi \cdot \nabla \sin\phi + \nabla \cdot (\mathbf{V}_\sigma \sin\phi) = 0. \quad (2.14)$$

The second term in (2.14) is not dominant because $\nabla \sin\phi$ is of order L/a , where a is the radius of the earth, and by assumption (e) $L/a \lesssim Ro$ for the motions of interest. $\sin\phi$ may therefore be considered a slowly varying function in the left-hand term of (2.13) and in the last term of (2.14). Hence (2.13) may be integrated to give

$$\psi = p / \sin\phi, \quad (2.15)$$

and (2.14) becomes

$$(\partial / \partial t + \mathbf{V}_\psi \cdot \nabla) (\nabla^2 \psi + Ro^{-1} \sin\phi) + \sin\phi \nabla \cdot \mathbf{V}_\sigma = 0. \quad (2.16)$$

The non-dimensional form of the continuity equation is

$$\kappa \epsilon \rho_s^* (\partial / \partial t + \mathbf{V} \cdot \nabla) \rho + \rho_s^* (1 + \kappa \epsilon Ro \rho) \nabla \cdot \mathbf{V}_\sigma + \partial [\rho_s^* (1 + \kappa \epsilon Ro \rho) w] / \partial z = 0, \quad (2.17)$$

which reduces to

$$\rho_s^* \nabla \cdot \mathbf{V}_\sigma + \partial (\rho_s^* w) / \partial z = 0 \quad (2.18)$$

in the first approximation.

On the assumption that the flow is adiabatic the first law of thermodynamics may be written [cf. Eq (2.1)]

$$(\partial / \partial t + \mathbf{V}^* \cdot \nabla^*) (\gamma^{-1} \ln p^* + \ln \alpha^*) + w^* \partial \ln \theta^* / \partial z^* = 0. \quad (2.19)$$

In non-dimensional form this becomes

$$\frac{[\partial / \partial t + (\mathbf{V}_\psi + Ro \mathbf{V}_\sigma) \cdot \nabla] (\mu \gamma^{-1} p)}{1 + \mu \kappa \epsilon Ro p} + \frac{[\partial / \partial t + (\mathbf{V}_\psi + Ro \mathbf{V}_\sigma) \cdot \nabla] \alpha}{1 + \kappa \epsilon Ro \alpha} + w \left[\frac{1}{\epsilon(z)} + Ro \frac{\partial \theta}{\partial z} \right] = 0. \quad (2.20)$$

From the definitions (2.6) we have

$$\theta \approx \mu \gamma^{-1} p + \alpha, \quad (2.21)$$

whence (2.20) reduces to

$$(\partial / \partial t + \mathbf{V}_\psi \cdot \nabla) \theta + \epsilon^{-1} w = 0, \quad (2.22)$$

in the first approximation.² Utilizing Eq (2.21), (2.1), (2.6), (2.9), (2.10) and (2.15) we find

$$\theta = \left[\frac{gD}{\gamma p_s^* \alpha_s^*} \psi + \alpha_s^* \frac{\partial}{\partial z} \left(\frac{\psi}{\alpha_s^*} \right) \right] \sin\phi = \left(\frac{\partial \psi}{\partial z} - \kappa \psi \right) \sin\phi \approx \frac{\partial \psi}{\partial z} \sin\phi, \quad (2.23)$$

so that (2.22) takes the form

$$\left(\frac{\partial}{\partial t} + \mathbf{V}_\psi \cdot \nabla \right) \frac{\partial \psi}{\partial z} + \frac{w}{\epsilon \sin\phi} = 0. \quad (2.24)$$

² It is not necessary to consider the two cases, $\epsilon \sim 1$ and $\epsilon \gg 1$, separately since the latter is an asymptotic limit of the former. Strictly, $\epsilon(z)$ is to be considered a slowly varying function of y , analogous to $\sin\phi$ in (2.15) and (2.16).

Elimination of w and $\mathbf{V}_\sigma$ from (2.16), (2.18) and (2.24) then gives

$$\left. \begin{aligned} (\partial/\partial t + \mathbf{V}_\psi \cdot \nabla)q &= 0, \\ q &\equiv \nabla^2 \psi + Ro^{-1} \sin \phi + \frac{\sin^2 \phi}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \psi}{\partial z} \right) \end{aligned} \right\} \quad (2.25)$$

This is the form taken by the theorem of conservation of potential vorticity (Rossby, 1940; Ertel, 1942) for quasi-geostrophic flows obeying the assumptions (2.5). To show this we introduce a quasi-Lagrangian coordinate system in which z is replaced by θ^* as vertical coordinate. Let differentiation at constant θ^* be distinguished from differentiation at constant z^* by the subscripts " θ " and " z ," respectively. For quasi-hydrostatic flow the potential vorticity Q in the θ^* -system may be written (Starr, 1945)

$$Q \equiv [\mathbf{k} \cdot (\nabla \times \mathbf{V})_\theta + Ro^{-1} \sin \phi] [\alpha^* \div \partial z / \partial \theta], \quad (2.26)$$

and the statement that Q is conserved,

$$(\partial/\partial t + \mathbf{V} \cdot \nabla)_\theta Q = 0. \quad (2.27)$$

Transforming to the z -system by means of the formulae,

$$\left(\frac{\partial}{\partial t} \right)_\theta \equiv \left(\frac{\partial}{\partial t} \right)_z - \left[\left(\frac{\partial}{\partial t} \right)_z \theta^* \right] \left(\frac{\partial \theta^*}{\partial z} \right)^{-1} \frac{\partial}{\partial z}, \quad (2.28)$$

we find

$$\left(\frac{\partial}{\partial t} \right)_\theta \equiv \left(\frac{\partial}{\partial t} \right)_z - \epsilon Ro \left[\left(\frac{\partial}{\partial t} \right)_z \theta \right] \frac{\partial}{\partial z}, \quad (2.29)$$

and

$$Q \approx \alpha_s^* \theta_s^* \kappa [(\nabla^2 \psi)_z + Ro^{-1} \sin \phi + \epsilon \sin \phi \partial \theta / \partial z] \quad (2.30)$$

correct to the first order in small quantities. Substitution of (2.30) in (2.29) then gives

$$\left(\frac{\partial}{\partial t} \right)_\theta Q \approx \alpha_s^* \theta_s^* \kappa \left(\frac{\partial}{\partial t} \right)_z q, \quad (2.31)$$

and

$$(\partial/\partial t + \mathbf{V} \cdot \nabla)_\theta Q \approx \alpha_s^* \theta_s^* \kappa (\partial/\partial t + \mathbf{V}_\psi \cdot \nabla)_z q = 0,$$

whence we rederive Eq (2.25).

To obtain the corresponding equation for an incompressible fluid one need only replace the potential temperature by the specific volume; this is accomplished by setting $\gamma = \infty$. The forms of (2.24) and (2.25) are seen to remain unchanged.

The condition to be satisfied at a rigid horizontal boundary results from setting $w=0$ in (2.24):

$$(\partial/\partial t + \mathbf{V}_\psi \cdot \nabla)(\partial \psi / \partial z) = 0. \quad (2.32)$$

At the outer boundary of an infinite compressible atmosphere the kinetic energy density, $\frac{1}{2} \rho_s^* |\nabla \psi|^2$ must be bounded. If the structure of the very high atmosphere is such that all energy generated in the lower atmosphere is reflected, the kinetic energy density and the vertical energy flux must approach zero. (See Charney and Drazin, 1961). We shall make the assumption that this is so in dealing with internal jet perturbations.

3. The stability of the internal jet

Let λ be the longitude and take $L=a$, the radius of the earth. (L is no longer to be regarded as a characteristic scale parameter.) If mean flow quantities are denoted by bars and perturbation quantities by primes, the perturbation forms of (2.24) and (2.25) become

$$\mathfrak{D}(\partial \psi' / \partial z) + v' \partial^2 \bar{\psi} / \partial \phi \partial z + (\epsilon \sin \phi)^{-1} w' = 0, \quad (3.1)$$

$$\mathfrak{D}q' + v' \partial \bar{q} / \partial \phi = 0, \quad (3.2)$$

where

$$\mathfrak{D} = \frac{\partial}{\partial t} + \bar{\omega} \frac{\partial}{\partial \lambda}, \quad v' = \frac{1}{\cos \phi} \frac{\partial \psi'}{\partial \lambda}, \quad (3.3)$$

$$q' = \nabla^2 \psi' + \frac{\sin^2 \phi}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \psi'}{\partial z} \right), \quad (3.4)$$

and $\bar{\omega} = -(\cos \phi)^{-1} \partial \bar{\psi} / \partial \phi$ is the non-dimensional angular velocity of the mean zonal flow. We suppose that the disturbances are of the form

$$\psi' = \Psi(\phi, z) e^{i(m\lambda + \nu t)}$$

and substitute into (3.1) and (3.2), obtaining

$$w' = -i \epsilon \sin \phi \left[(\nu + m \bar{\omega}) \frac{\partial \Psi}{\partial z} - m \frac{\partial \bar{\omega}}{\partial z} \Psi \right] e^{i(m\lambda + \nu t)}, \quad (3.5)$$

$$\begin{aligned} \frac{1}{\cos \phi} \frac{\partial}{\partial \phi} \left(\cos \phi \frac{\partial \Psi}{\partial \phi} \right) + \frac{\sin^2 \phi}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \Psi}{\partial z} \right) \\ - \frac{m^2}{\cos^2 \phi} \Psi + \frac{m}{\nu + m \bar{\omega}} \frac{\partial \bar{q}}{\partial \phi} \frac{\Psi}{\cos \phi} = 0. \end{aligned} \quad (3.6)$$

Let us assume that the flow is unstable so that $\nu = \nu_r + i\nu_i$ with $\nu_i < 0$. The potential temperature gradient vanishes at a rigid horizontal boundary:

$$\frac{\partial \bar{\theta}}{\partial \phi} = \sin \phi \frac{\partial^2 \bar{\psi}}{\partial \phi \partial z} = \sin \phi \cos \phi \frac{\partial \bar{\omega}}{\partial z} = 0. \quad (3.7)$$

Eq (3.5) then shows that at this boundary,

$$\partial \Psi / \partial z = 0. \quad (3.8)$$

Or, if one or both of the horizontal boundaries is at infinity, the vanishing of the kinetic energy density

and the vertical energy flow imply:

$$\lim_{z \rightarrow \infty} \rho_s^* \psi' w' = 0, \quad \lim_{z \rightarrow \infty} \rho_s^* (\nabla \psi')^2 = 0. \quad (3.9)$$

Multiplication of (3.5) by Ψ_* , the complex conjugate of Ψ , then gives

$$\lim_{z \rightarrow \infty} \epsilon \rho_s^* \Psi_* \partial \Psi / \partial z = 0, \quad (3.10)$$

providing $\rho_s^* \bar{\omega}$ is bounded, as we may assume.

Multiplication of (3.6) by $\rho_s^* \Psi_* \cos \phi d\phi dz$ and integration between the horizontal boundaries, $z = z'$ and z'' , of the fluid now gives

$$\begin{aligned} & \int \int \rho_s^* \left(\cos \left| \frac{\partial \Psi}{\partial \phi} \right|^2 + \epsilon \sin^2 \phi \cos \phi \left| \frac{\partial \Psi}{\partial z} \right|^2 \right. \\ & \quad \left. + \frac{m^2}{\cos \phi} |\Psi|^2 - \frac{m}{\nu + m\bar{\omega}} \frac{\partial \bar{q}}{\partial \phi} |\Psi|^2 \right) d\phi dz \\ & = - \int \epsilon \rho_s^* \sin^2 \phi \cos \phi \Psi_* \frac{\partial \Psi}{\partial z} d\phi \Big|_{z'}^{z''} = 0 \end{aligned} \quad (3.11)$$

in virtue of (3.8) or (3.10). As both the real and the imaginary parts of the left-hand side must vanish, we must have

$$\nu_i \int \int \frac{\rho_s^*}{|\nu + m\bar{\omega}|^2} \frac{\partial \bar{q}}{\partial \phi} |\Psi|^2 d\phi dz = 0,$$

but this can be so only if $\partial \bar{q} / \partial \phi$ vanishes along some surface within the fluid. If it does not vanish, ν_i cannot be different from zero and the disturbance cannot be unstable. Hence, in view of (2.31), we may state the general theorem:

Zonal flow in a stratified rotating atmosphere which (a) is bounded by rigid horizontal boundaries, or (b) extends to infinity, is stable with respect to axially asymmetric disturbances if the gradient of potential vorticity in isentropic surfaces does not vanish and (a) the potential temperature is constant at the rigid boundaries, or (b) perturbation energy is reflected at infinity.

The vanishing of the potential vorticity gradient is thus a sufficient condition for stability and a necessary condition for instability.

A physical interpretation of the stability criterion similar to one advanced by Lin (1945) for plane parallel flow in an inviscid fluid will now be given. The argument differs from Lin's only in the avoidance of the use of Green's functions.

Eq (3.2) is multiplied by v' to give

$$q' \mathcal{D}v' = v'^2 \partial \bar{q} / \partial \phi + \mathcal{D}(q'v'),$$

from which we derive, through multiplication by

$\rho_s^* \cos \phi$ and integration over the entire fluid,

$$\begin{aligned} \int_{\tau} \rho_s^* q' \mathcal{D}v' \cos \phi d\tau &= \int_{\tau} \rho_s^* v'^2 (\partial \bar{q} / \partial \phi) \cos \phi d\tau \\ &+ d/dt \int_{\tau} \rho_s^* v' q' \cos \phi d\tau. \end{aligned} \quad (3.12)$$

Here τ denotes the fluid volume and $d\tau$ the volume element $\cos \phi d\lambda d\phi dz$. The quantity $v'q'$ is written

$$\begin{aligned} v'q' &= \frac{1}{\cos \phi} \frac{\partial \psi'}{\partial \lambda} \left[\nabla \cdot \nabla \psi' + \frac{\sin^2 \phi}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \psi'}{\partial z} \right) \right] \\ &= \frac{1}{\cos \phi} \left[\nabla \cdot \left(\frac{\partial \psi'}{\partial \lambda} \nabla \psi' \right) - \frac{1}{2} \frac{\partial}{\partial \lambda} (\nabla \psi' \cdot \nabla \psi') \right. \\ & \quad \left. + \epsilon \sin^2 \phi \left(\frac{\partial \psi'}{\partial z} \right)^2 \right] + \frac{\sin^2 \phi}{\rho_s^* \cos \phi} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \psi'}{\partial \lambda} \frac{\partial \psi'}{\partial z} \right). \end{aligned}$$

If the flow is bounded by a vertical wall at some latitude, or if the integration extends over the entire sphere, only the last term in this expression for $v'q'$ contributes to the second integral on the right-hand side of (3.12), which then reduces to the surface integral,

$$- \frac{d}{dt} \int \epsilon \rho_s^* \frac{\partial \psi'}{\partial \lambda} \frac{\partial \psi'}{\partial z} \sin^2 \phi \cos \phi d\lambda d\phi \Big|_{z'}^{z''}. \quad (3.13)$$

If the potential temperature gradient vanishes at rigid horizontal boundaries, or if one or more of the boundaries extends to infinity, it follows from (3.1) and (3.9) that the integrand in (3.13) must vanish. Hence (3.13) is zero and (3.12) becomes

$$\int_{\tau} \rho_s^* q' \mathcal{D}v' \cos \phi d\tau = \int_{\tau} \rho_s^* v'^2 \partial \bar{q} / \partial \phi \cos \phi d\tau. \quad (3.14)$$

Since $\mathcal{D}v'$ is the perturbation acceleration in the meridional direction, this result may be given the following physical interpretation: Let us imagine that an infinitesimal disturbance is imparted to the fluid such that a parcel of fluid is displaced from the zonal circle ϕ_1, z_1 to the zonal circle ϕ_2, z_2 in the time Δt . Because of the conservation of q , expressed by (2.25), the perturbation in q resulting from the displacement is given by

$$q'(\phi_2, z_2) = \bar{q}(\phi_1, z_1) - \bar{q}(\phi_2, z_1) \approx -\partial \bar{q} / \partial \phi v' \Delta t.$$

If $\bar{q}$ is a monotonic function of ϕ , say for definiteness an increasing function, then q' will be positive or negative according as v' is negative or positive. Hence, by (3.14), the meridional acceleration $\mathcal{D}v'$ will be of the opposite sign to v' , the meridional displacement velocity. The same conclusion is reached if $\bar{q}$ is a

monotonic decreasing function of ϕ . Thus we derive the same result as before: the condition $\partial\bar{q}/\partial\phi \neq 0$ is a sufficient condition for instability. If $\partial\bar{q}/\partial\phi$ vanishes, the restoring force also vanishes, and the possibility exists that a disturbance may grow. Again, the condition of vanishing potential vorticity gradient is a necessary but not a sufficient condition for instability.

Our progress in establishing sufficient conditions is indicated in the appendix, where we extend the sufficiency conditions for the classical jet to a three-dimensional flow. These conditions are, however, too special for application in the next section to the polar-night jet.

4. The stability of the polar-night jet

Using North American data, Krishnamurti (1959) computed vertical cross-sections of $\bar{\theta}^*$, $\bar{u}^*$ ($=\bar{\omega}a \cos\phi$), and $\bar{Q}^*$ through the polar-night jet for the period 31 December 1957 to 3 January 1958. He oriented his data with respect to the jet axis in order to eliminate the fictional smoothing introduced by taking zonal averages when the motion is not precisely zonal. In

the computation of the potential vorticity,

$$\bar{Q}^* = - \left[2\Omega \sin\phi - \frac{1}{a \cos\phi} \frac{\partial}{\partial\phi} (\bar{u}^* \cos\phi) \right] g \frac{\partial\bar{\theta}^*}{\partial p^*}, \quad (4.1)$$

he defined $\bar{u}^*$ as the wind component parallel to the jet axis. A cross-section of $\bar{\theta}^*$, $\bar{u}^*$ and $\bar{Q}^*$ is shown in Fig. 1. There is only a slight indication of a maximum of $\bar{Q}^*$ along isentropic surfaces in the northernmost and uppermost part of the cross-section. However, one must take into account that Krishnamurti's zonal velocities were computed, not geostrophically from pressure data, but from radar wind measurements which often cannot be made when the winds are so strong at high levels that the balloon drifts very low on the horizon. Since the effect is to underestimate the wind intensity in strong currents, separate cross-sections of $\bar{\theta}^*$, $\bar{u}^*$ and $\bar{Q}^*$ were computed by Mr. C. Leovy for 14 October, 1200 GMT and 25 December, 1200 GMT, 1957 at about 150W. These are shown in Figs. 2 and 3. We see that at the earlier date when the jet has begun to form but has not reached its maximum intensity there is little or no maximum and a weak minimum in

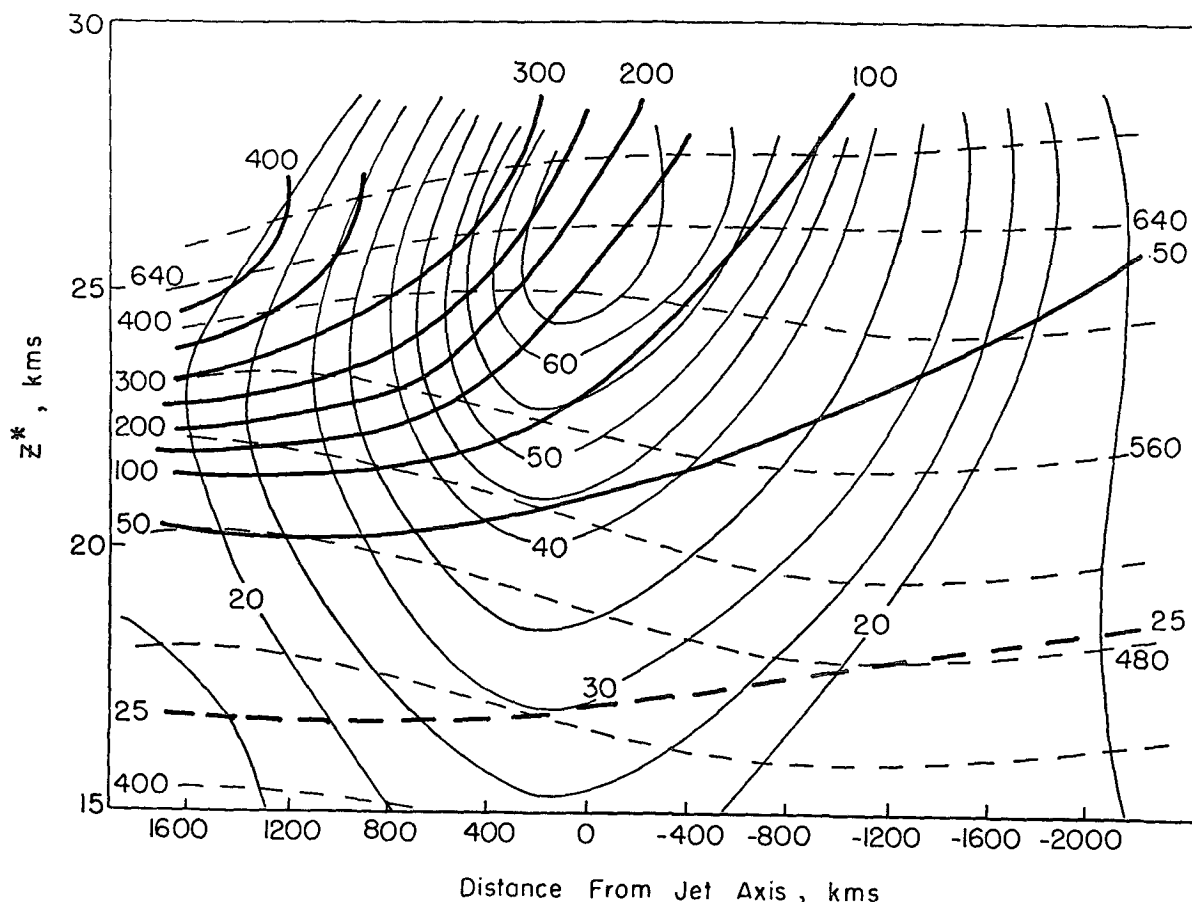


FIG. 1. Vertical cross-section of westerly wind component in m/sec (thin solid lines), potential temperature in deg K (dashed lines) and potential vorticity in 10^{-2} cm/sec²/mb/deg (heavy solid lines). The data are averaged over approximately a 100-deg longitude interval covering North America for the four-day period, 31 December 1957 to 3 January 1958. (After Krishnamurti.)

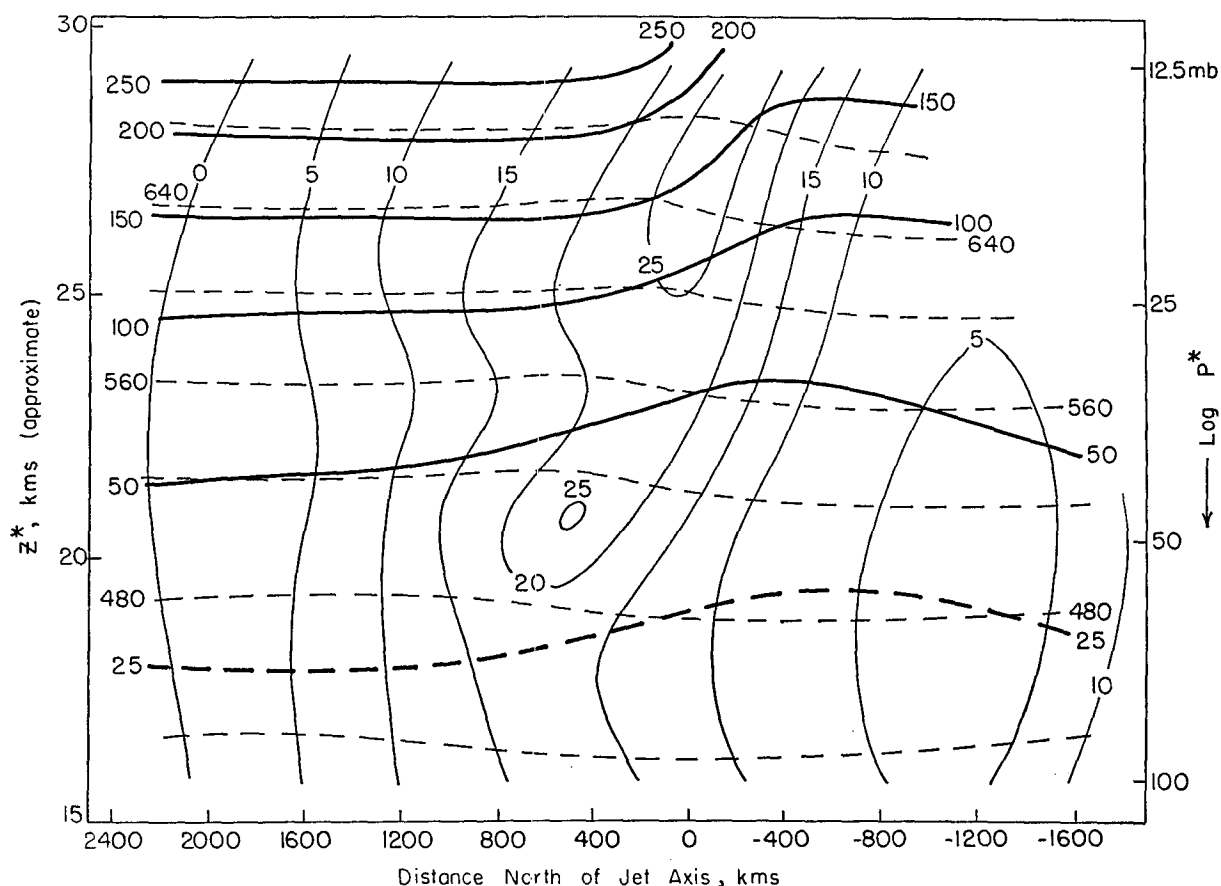


FIG. 2. Vertical cross-section of westerly wind component (thin solid lines), potential temperature (dashed lines) and potential vorticity (heavy solid lines) at approximately 150W longitude on 1200 GMT, 14 October 1957. Units as in Fig. 1.

$\bar{Q}^*$ along isentropic surfaces. However, at the later date, when the jet is near its maximum intensity, there are distinct maxima and minima in $\bar{Q}^*$ from 16 km to the highest level at which data are available.

Leovy's observations point to high level instability of the internal jet type as a possible cause of the meandering of the jet in the fall and its final breakdown in late winter. If this were so, the amplification of the Gulf of Alaska anticyclone, which appears to precede the breakdown, could be interpreted as the unstable growth of a stationary finite-amplitude perturbation initiated by small geographically fixed influences, possibly topographical or radiative, but deriving its energy principally from the mean zonal flow.³

Evidence from mean cross-sections compiled by Murgatroyd (1957) and Batten (1961) show that in midwinter the jet intensity, and therefore the maximum in the potential vorticity gradient, increases upward to about 60 km and then decreases. This suggests that an unstable perturbation would form first in the vicinity of 60 km and then propagate downward below 60 km

and upward above 60 km. In the region below 30 km there is evidence (Craig and Hering, 1959) that the disturbances associated with the breakdown of the jet do in fact propagate downwards.

The instability hypothesis is also supported by the fact that both observation and theory (Charney and Drazin, *loc. cit.*) indicate little coupling between tropospheric and upper stratospheric motions so that the seat of the perturbation energy must lie in the upper stratosphere itself.

However, it must be emphasized that no definite conclusion as to the existence of instability can be reached until it has been demonstrated that the jet-stream satisfies sufficient as well as necessary conditions for instability and that the unstable perturbations predicted by theory do in fact resemble the observed perturbations. Preliminary calculations performed by Mr. Leovy indicate that the necessary condition for instability may also be satisfied by the weaker summer stratospheric easterlies where no sudden breakdowns occur. It is likely that this condition is also satisfied in the southern hemisphere polar-night jet, where no comparable breakdown phenomena have yet been observed to occur before the return of the sun to the polar

³ The perturbation motion might also alter the photochemical equilibrium and the radiative transfer properties of the ozone layer in such a way as to feed energy back into the perturbation.

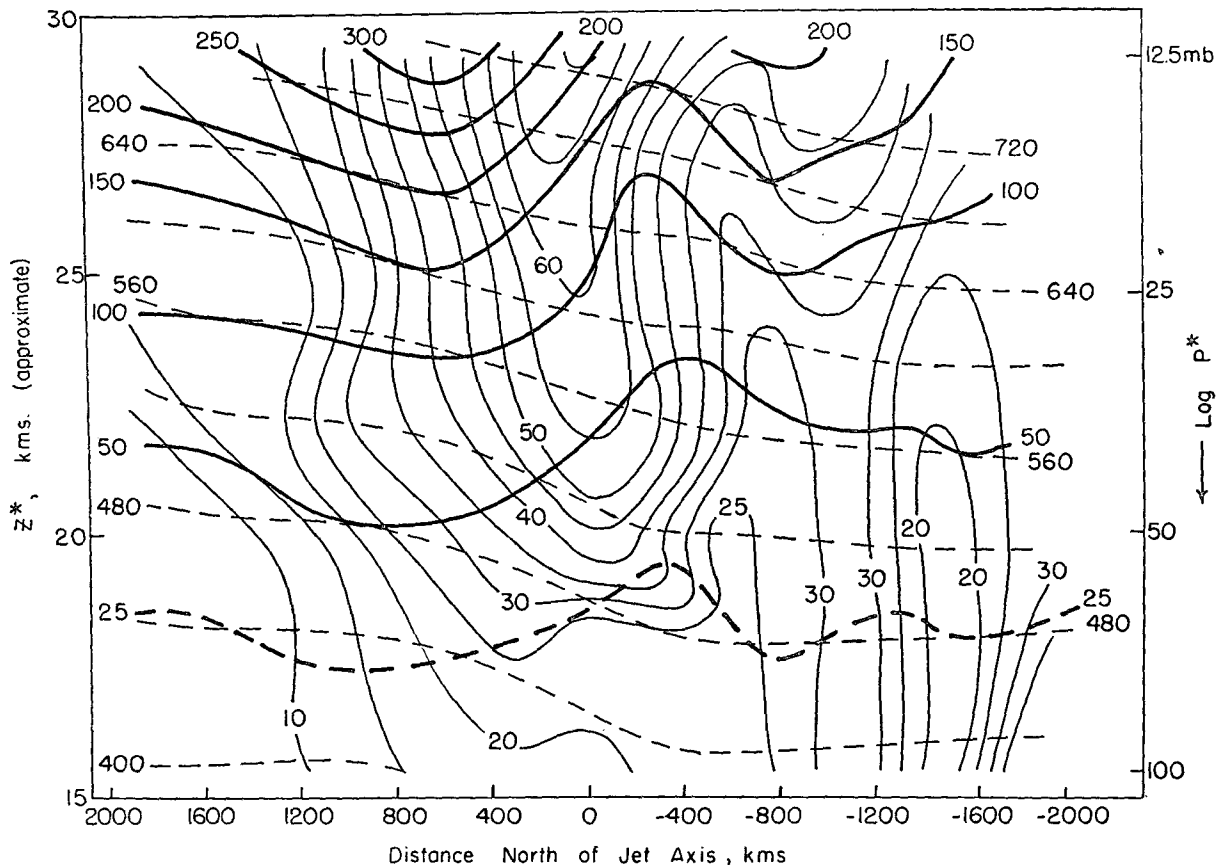


FIG. 3. Vertical cross-section of westerly wind component (thin solid lines), potential temperature (dashed lines) and potential vorticity (heavy solid lines) at approximately 150W longitude on 1200 GMT, 25 December 1957. Units as in Fig. 1.

regions destroys the circumpolar cyclone (H. Wexler, private communication).

5. The stabilizing effect of the surface boundary condition

All of the investigators of the baroclinic stability problem mentioned in the introduction to this article have found that a zonal current with a uniform horizontal temperature gradient is unstable. The source of the energy for the growing disturbances is the potential and internal energy of the mean flow associated with the horizontal temperature gradients. The internal jets discussed in section 3 of this article are permitted to have strong horizontal temperature gradients and yet may be absolutely stable. What is the stabilizing factor? Formally, of course, the difference lies in the condition that the potential temperature θ is constant at a rigid horizontal boundary, implying that $(\partial\psi'/\partial\lambda)(\partial\psi'/\partial z)=0$ in (3.13) and $\partial\psi/\partial z=0$ in (3.11). It may be shown that the former quantity is proportional to the horizontal eddy flux of heat and determines the conversion of mean flow potential and internal energy into per-

turbation potential and internal energy just as the Reynolds stress determines the conversion of mean flow kinetic energy into perturbation kinetic energy. The energy equation for motions governed by Eq (2.25) are derived as follows. Write $\psi=\psi+\psi'$, $w=w'$, multiply (2.16) by $-\rho_s^*\psi'$, (2.24) by $\rho_s^*\epsilon \sin^2\phi(\partial\psi'/\partial z)$, add and integrate with respect to volume from the ground to the level z . Integration by parts and application of Gauss' theorem then gives

$$\begin{aligned} \frac{d}{dt} \int_{\tau} \left[\frac{\rho_s^*}{2} (\nabla\psi')^2 + \frac{\rho_s^*}{2} \epsilon \sin^2\phi \left(\frac{\partial\psi'}{\partial z} \right)^2 \right] d\tau \\ = \int_{\tau} \frac{\rho_s^*}{\cos\phi} \frac{\partial\psi'}{\partial\lambda} \frac{\partial\psi'}{\partial\phi} \frac{\partial(\bar{\omega} \cos\phi)}{\partial\phi} d\tau \\ + \int_{\tau} \frac{\rho_s^* \epsilon \sin^2\phi}{\cos\phi} \frac{\partial\psi'}{\partial\lambda} \frac{\partial\psi'}{\partial z} \frac{\partial(\bar{\omega} \cos\phi)}{\partial z} d\tau \\ - \int_S \rho_s^* p' w' dS, \quad (5.1) \end{aligned}$$

where τ denotes the volume between $z=0$ and $z=\text{constant}$, and S is the plane, $z=\text{constant}$. The first term in the integral on the left-hand side is the perturbation kinetic energy and the second the perturbation "available" potential energy (Lorenz, 1955). The first integral on the right is the rate of conversion of mean flow kinetic energy into perturbation kinetic energy and the second the rate of conversion of mean flow available potential energy into perturbation available potential energy. The last term is the vertical flow of energy, which is assumed to approach zero with increasing height.

The form of Eq (2.25) and the correspondence between the energy transfer terms in (5.1) suggests a correspondence between the problem of baroclinic stability and that of two-dimensional parallel flow. We shall show that such a correspondence exists and that the stabilizing effect of the internal jet boundary condition has a physical counterpart in the two-dimensional analogue.

Consider for simplicity a rotating cylindrical annulus whose inner and outer radii, R_1 and R_2 satisfy the inequality $(R_2 - R_1)/R_1 \ll 1$, so that Cartesian coordinates, x and y , may be introduced for the zonal and meridional directions. On the assumption that the scaling assumptions (2.5) continue to hold, the equations of motion are obtained simply by setting $\sin\phi = 1$ in (2.15), (2.16) and (2.24). Elimination of w with the aid of (2.18) then gives

$$(\partial/\partial t + \mathbf{V}_\psi \cdot \nabla)(\partial\psi/\partial z) + \epsilon^{-1}w = 0, \quad (5.2)$$

and

$$\begin{aligned} &(\partial/\partial t + \mathbf{V}_\psi \cdot \nabla)q = 0, \\ &q \equiv \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{1}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial\psi}{\partial z} \right). \end{aligned} \quad (5.3)$$

Substituting

$$\psi' = \Psi(y, z) e^{ik(x - ct)}$$

into (5.3), we get

$$\frac{\partial^2\Psi}{\partial y^2} + \frac{1}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial\Psi}{\partial z} \right) + \left(-k^2 + \frac{\partial\bar{q}/\partial y}{\bar{u} - c} \right) \Psi = 0, \quad (5.4)$$

with

$$\frac{\partial\bar{q}}{\partial y} = -\frac{\partial^2\bar{u}}{\partial y^2} - \frac{1}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial\bar{u}}{\partial z} \right). \quad (5.5)$$

The surface boundary condition is obtained by setting $w=0$ in (5.2). This gives

$$(\bar{u} - c) \partial\Psi/\partial z - (\partial\bar{u}/\partial z) \Psi = 0. \quad (5.6)$$

In the pure baroclinic stability problem $\bar{u}$ is not a function of y . Hence it may be assumed that Ψ is independent of y , or else the y variation may be separated with the factor $e^{i\ell y}$. In either case the eigenfunction equation has the form

$$\frac{1}{\rho_s^*} \frac{d}{dz} \left(\epsilon \rho_s^* \frac{d\Psi}{dz} \right) + \left[-k^2 + \frac{-\frac{1}{\rho_s^*} \frac{d}{dz} \left(\epsilon \rho_s^* \frac{d\bar{u}}{dz} \right)}{(\bar{u} - c)} \right] \Psi = 0. \quad (5.7)$$

Let us for the moment make the assumption that ϵ is constant. Introducing the vertical coordinate transformation $\xi = \epsilon^{1/2}z$, Eq (5.7) and (5.6) become

$$\frac{1}{\rho_s^*} \frac{d}{d\xi} \left(\rho_s^* \frac{d\Psi}{d\xi} \right) + \left[-k^2 + \frac{-\frac{1}{\rho_s^*} \frac{d}{d\xi} \left(\rho_s^* \frac{d\bar{u}}{d\xi} \right)}{\bar{u} - c} \right] \Psi = 0, \quad (5.8)$$

$$(\bar{u} - c) d\Psi/d\xi - (d\bar{u}/d\xi) \Psi = 0. \quad (5.9)$$

For comparison we consider the equation for small perturbations of the two-dimensional zonal flow of a stratified incompressible fluid in the absence of gravity. Introducing the stream function: $\mathbf{V} = \mathbf{k} \times \nabla\chi$, and taking the curl of the equation of motion,

$$\rho^* (\partial/\partial t + \mathbf{V} \cdot \nabla) \mathbf{V} = -U^{-2} \nabla p^*,$$

we get

$$(\partial/\partial t + \mathbf{V} \cdot \nabla) \nabla^2 \chi + (\nabla \rho^* / \rho^*) \cdot (\partial/\partial t + \mathbf{V} \cdot \nabla) \nabla \chi = 0.$$

Substitution of $\chi' = X(y) e^{ik(x - ct)}$ and neglect of second order quantities then gives

$$\frac{1}{\bar{\rho}^*} \frac{d}{dy} \left(\bar{\rho}^* \frac{dX}{dy} \right) + \left[-k^2 + \frac{-\frac{1}{\bar{\rho}^*} \frac{d}{dy} \left(\bar{\rho}^* \frac{\partial\bar{u}}{\partial y} \right)}{\bar{u} - c} \right] X = 0, \quad (5.10)$$

which is identical with (5.8) if X is identified with Ψ , y with ξ , and $\bar{\rho}^*$ with $\bar{\rho}_s^*$.

A number of interesting consequences may be deduced from this isomorphism. Thus if $\rho_s^* \equiv 1$ and $\bar{u}$ is a linear function of ξ , one finds that Eady's (1949) stability problem reduces to that of two-dimensional inviscid Couette flow (linear shear in a homogeneous incompressible fluid). Now it is well-known that Couette flow is stable if confined between rigid boundaries or if extended laterally to infinity, yet Eady has shown that the corresponding baroclinic flow is unstable when the wavelength is sufficiently large. The difference lies in the boundary conditions. A rigid boundary in Couette flow corresponds to a boundary at which Ψ vanishes in the baroclinic analogue, not to a boundary at which

(5.9) holds. The latter condition corresponds to a *free* boundary in Couette flow. Hence, by analogy: Couette flow with free boundaries is unstable for sufficiently large perturbation wavelengths. To show that (5.9) is in fact the free boundary condition, we write the condition that the pressure must vanish at a free boundary:

$$p = \bar{p} + p' = p' = 0.$$

The x -component of the equation of motion at the boundary then becomes

$$\left(-\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x}\right) \frac{\partial \chi'}{\partial y} + \frac{\partial \chi'}{\partial x} \frac{\partial \bar{u}}{\partial y} - \frac{1}{\bar{\rho}^*} \frac{\partial p'}{\partial x} = 0,$$

which reduces to

$$(\bar{u} - c) \partial X / \partial y - (\partial \bar{u} / \partial y) X = 0$$

on substitution of $\chi' = X(y)e^{ik(x-ct)}$. This equation is identical to (5.9) if y is again identified with ξ .

Available potential energy in baroclinic flow corresponds to what may be called "available kinetic energy" in two-dimensional flow. (The mean velocity in shearing flow can always be altered by virtual displacements which conserve total mass and momentum so as to decrease the mean flow kinetic energy.) It is known, however, that the mere existence of available kinetic energy does not insure that the dynamical constraints will permit its release, as in the case of Couette flow with rigid boundaries. The vanishing heat flow boundary condition prevents the release of available potential energy in a similar manner. This may be shown by slightly generalizing a theorem of Foote and Lin (1950) pertaining to the variation of the Reynolds stress in two-dimensional flow.

Write for the real part of the eddy flux of heat

$$\begin{aligned} B &= \rho_s^* \epsilon Re \frac{\partial \psi'}{\partial x} Re \frac{\partial \psi'}{\partial z} \\ &= \rho_s^* \epsilon \frac{ik}{2} \left[\overline{\Psi e^{ik(x-ct)} - \Psi_* e^{-ik(x-c_*t)}} \right] \\ &\quad \times \frac{1}{2} \left[\frac{d\Psi}{dz} e^{ik(x-ct)} + \frac{d\Psi_*}{dz} e^{-ik(x-c_*t)} \right] \\ &= \rho_s^* \epsilon \frac{ik}{4} \left(\Psi \frac{d\Psi_*}{dz} - \frac{d\Psi}{dz} \Psi_* \right) e^{2kc_i t}, \end{aligned}$$

where c_i is the imaginary part of the phase velocity c . Multiply (5.7) and its conjugate by $\rho_s^* \Psi_*$ and $\rho_s^* \Psi$,

respectively, and then subtract. One obtains

$$\frac{d}{dz} \left[\rho_s^* \epsilon \left(\Psi_* \frac{d\Psi}{dz} - \Psi \frac{d\Psi_*}{dz} \right) \right] = 2ic_i \rho_s^* \frac{|\Psi|^2}{|\bar{u} - c|^2} \frac{\partial \bar{q}}{\partial y},$$

or

$$\frac{dB}{dz} = -\frac{kc_i}{2} \rho_s^* \frac{\partial \bar{q}}{\partial y} \frac{|\Psi|^2}{|\bar{u} - c|^2} e^{2kc_i t}. \quad (5.11)$$

Suppose now that $\partial \bar{q} / \partial y \neq 0$ everywhere in the fluid and that B vanishes at lower and upper horizontal boundaries. It then follows from (5.11) that $c_i = 0$ or $\Psi = 0$, i.e., the wave is neutral or non-existent. If B does not vanish at the lower boundary or at infinity, no such statement can be made.

Eq (5.11) can be extended to flows on a rotating earth with vertical shearing discontinuities by introducing the so-called Rossby β -plane approximation. Since the scale of the disturbance of the zonal flow is assumed to be such that $\sin \phi$ is a slowly varying function in (2.25), we shall be justified in replacing the factor $\sin^2 \phi$ by $\sin^2 \phi_0$, corresponding to the mean latitude ϕ_0 , and in setting $Ro^{-1} \sin \phi = \beta y$, where y is the distance measured northward along a meridian from the latitude ϕ_0 . Introducing the zonal coordinate x , measured eastward from a fixed meridian, we ignore the convergence of meridians and assume that x, y, z is a Cartesian coordinate system. Thus

$$q = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \beta y + \frac{1}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial}{\partial z} \right),$$

and

$$\frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{1}{\rho_s^*} \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \bar{u}}{\partial z} \right),$$

with $\partial^2 \bar{u} / \partial y^2 = 0$ in the pure baroclinic stability problem. With the above definition of q one proceeds as before to derive (5.11).

In Murray's problem $\partial \bar{q} / \partial y > 0$ everywhere and $B = 0$ at the boundaries but is discontinuous at the tropopause, where there is a jump in ϵ and $\partial \bar{u} / \partial z$. The jump in B is obtained from the perturbation form of (2.24) and the jump conditions $[w'] = [\psi'] = [\bar{u}] = [\rho_s^*] = 0$ at the tropopause, $z = z_T$ (Charney and Drazin, *loc. cit.*). Thus we find

$$\left[\frac{\partial \Psi}{\partial z} \right] = \left[\frac{\partial \bar{u}}{\partial z} \right] \frac{\Psi}{\bar{u} - c},$$

$$[B] = \rho_s^* \frac{kc_i}{2} \frac{|\Psi|^2}{|\bar{u} - c|^2} \left[\frac{\partial \bar{u}}{\partial z} \right] e^{2kc_i t}.$$

Integrating (5.11) from the lower boundary z_G to a level z above the tropopause discontinuity, we obtain

$$B(z) = \frac{kc_i}{2} e^{2kc_i z} \left\{ \left(\rho_s^* \left[\epsilon \frac{\partial \bar{u}}{\partial z} \right] \frac{|\Psi|^2}{|\bar{u} - c|^2} \right)_{z=z_T} + \int_{z_G}^z \rho_s^* \frac{\partial \bar{q}}{\partial y} \frac{|\Psi|^2}{|\bar{u} - c|^2} dz \right\}.$$

Since $[\epsilon \partial \bar{u} / \partial z]$ and $\partial \bar{q} / \partial y$ are positive in Murray's problem we may conclude that the quantity in braces is positive and hence c_i must vanish if the upper boundary condition $B=0$ is to be satisfied, i.e., his flow is stable. The stability argument in section 3 may be extended in a similar manner to include discontinuities.

If one does not assume that $B=0$ at the lower boundary, as in the baroclinic stability problem considered by Charney (1947), B need not vanish, and the zonal flow may be unstable. In Eady's baroclinic stability problem $\partial \bar{q} / \partial y$ is identically zero and we may conclude that B must be constant throughout the fluid but not necessarily zero. It follows from (5.1) that there will be a non-vanishing contribution to the growth of perturbation energy if B does not vanish at the boundaries. If the horizontal boundaries in Eady's model are removed to infinity and there is no energy flow at infinity, B must tend toward zero at infinity, and the flow is stable, a result already known for Couette flow.

The destabilizing effect of a horizontal temperature gradient at a rigid boundary in a baroclinic zonal flow where the isentropic potential vorticity gradient does not vanish may be likened to the destabilizing effect of "small" viscosity in Poiseuille flow, or in other two-dimensional flows with non-vanishing vorticity gradients and rigid walls. The temperature gradient permits a non-vanishing eddy flow of heat at a rigid boundary and this is analogous to the non-vanishing Reynolds stress close to a rigid boundary which is permitted by the action of viscosity.

The above considerations raise the question as to whether a "small" surface temperature gradient is capable of overcoming a monotonic variation in potential vorticity or, contrariwise, whether cyclogenesis can be inhibited unless and until the concentration of surface isotherms reaches a critical value which is necessary to overcome the dynamical constraints due to lateral shear. If the latter statement proves to be correct then we may reasonably speculate that low level fronts perform a similar role in the release of available energy by reducing the total constraint on a cyclonic system.

APPENDIX

A strong sufficient condition for instability

For several reasons, not the least of which is the inherent disparity between model and nature, the potential utility of an analysis like the present lies in suggesting sufficient rather than necessary conditions for the instability of a flow regime. The point is illustrated by the problem of shear flow in a channel where the Rayleigh inflection criterion is (in fact) neither necessary (as mentioned above) nor sufficient for instability. A theorem of Tollmien (Lin, 1955, p. 123) however, insures us that a certain class of *symmetrical* jets must break up spontaneously (in the limit of small viscosity).

In looking for an analogue of the Tollmien sufficiency theorem we note that the significance of the symmetry condition is related to the elimination of the singularities in the eigenfunction equation at the critical points, viz., where phase speed and current speed are equal. These critical points become a "critical curve" in the three dimensional case. Consider the class of free jets (no boundaries) which have the following properties for their mean fields. The jet has a maximum velocity at $(0,0)$ in the (y,z) plane, and the isopleths of current, $\bar{u}(y,z) = \text{constant}$, surround this center as simple closed curves. On each isentrope in the meridional section, find the points of maximum and minimum potential vorticity. A sufficient condition for instability is that the curve connecting these extrema coincides with some (any) velocity isopleth. The curve $\bar{u}(y,z) = \bar{u}_c$ for which this occurs is the critical curve and there are neutral disturbances which move with the phase speed $c = \bar{u}_c$. We then show that waves slightly longer than the neutral one have positive growth rate. Continuity arguments may be invoked to conclude that a small deformation of this mean field, such that the $\bar{u}$ and vorticity isopleth are no longer in exact coincidence, will still be unstable. A "weaker" (and more useful) sufficiency theorem would presumably supply a quantitative measure of the minimum critical distortion which is necessary for instability.

The analysis is based on the annulus approximation (5.4). We assume that $\partial \bar{q} / \partial y$ vanishes on the isopleth $\bar{u}(y,z) = \bar{u}_c$, or, more specifically,

$$0 \leq \frac{(\partial \bar{q} / \partial y)}{u(y,z) - u_c} \equiv G^2(y,z) < \infty^4. \quad (\text{A.1})$$

⁴ For example, any solution of

$$\frac{\partial^2 \bar{u}}{\partial y^2} + \frac{1}{\rho_s^*} \frac{\partial}{\partial z} \left(\rho_s^* \epsilon \frac{\partial \bar{u}}{\partial z} \right) + l^2 \bar{u} = -m^2, \\ \text{grad } \bar{u}(\pm \infty, \pm \infty) = 0,$$

where l^2 and m^2 are arbitrary positive numbers, is eligible for consideration here.

Define an integral (over all y, z) as follows:

$$I = \frac{\int \int \rho_s^* G^2(y, z) \Psi^2 dy dz - \int \int \rho_s^* \left(\frac{\partial \Psi}{\partial y} \right)^2 dy dz - \int \int \epsilon \rho_s^* \left(\frac{\partial \Psi}{\partial z} \right)^2 dy dz}{\int \int \rho_s^* \Psi^2 dy dz}. \quad (\text{A.2})$$

For any real, square integrable Ψ , $I < \text{maximum } G^2$. Secondly, there exists some Ψ (e.g., $\Psi = \exp - (y^2 + z^2)W^{-2}$, $W \rightarrow \infty$) which renders I positive. Thus (6.2) possesses an absolute positive maximum for a Ψ which satisfies the Euler-Lagrange equation of the variational problem. Hence there exists a solution of

$$\rho_s^*(z) \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \Psi}{\partial z} \right) + \rho_s^* G^2 \Psi - I_{\max} \rho_s^* \Psi = 0, \quad (\text{A.3})$$

$$\Psi(\pm \infty, \pm \infty) = 0.$$

This equation is similar to (5.4), and we conclude that the latter equation has a solution when $k^2 = I_{\max}$ and $c = u_c$.

It only remains to show that there are unstable solutions of (5.4) in the vicinity of the eigenvalue $k^2 = \max I$. Let $c_1 = a + ib$ be the complex phase speed of a disturbance Ψ_1 of wave number k_1 . As $k_1 \rightarrow k$, $c_1 \rightarrow u_c$, $\Psi_1 \rightarrow \Psi$. The equations satisfied by the neutral disturbance and its neighbor are:

$$\rho_s^* \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \Psi}{\partial z} \right) - k^2 \rho_s^* \Psi + \rho_s^* G^2 \Psi = 0, \quad (\text{A.4})$$

$$\rho_s^* \frac{\partial^2 \Psi_1}{\partial y^2} + \frac{\partial}{\partial z} \left(\epsilon \rho_s^* \frac{\partial \Psi_1}{\partial z} \right) - k_1^2 \rho_s^* \Psi_1 + \rho_s^* G^2 \frac{(\bar{u} - u_c)}{\bar{u} - c_1} \Psi_1 = 0. \quad (\text{A.5})$$

Multiply (6.4) by Ψ_1 , (6.5) by Ψ , integrate, and subtract the two equations to get:

$$(k_1^2 - k^2) \int \int \rho_s^* \Psi \Psi_1 dy dz + \int \int \rho_s^* G^2 \Psi \Psi_1 \left(\frac{u_c - c_1}{\bar{u} - c_1} \right) dy dz = 0. \quad (\text{A.6})$$

In the limit $dk^2 = k_1^2 - k^2$, $dc = c_1 - u_c$, $\Psi_1 \rightarrow \Psi$,

$$\frac{dk^2}{dc} = \left[\int \int \rho_s^* \Psi^2 dy dz \right]^{-1} \left[\lim_{\substack{b \rightarrow 0 \\ a \rightarrow u_c}} \int \int \rho_s^* G^2 \Psi^2 \frac{\bar{u} - a}{(\bar{u} - a)^2 + b^2} dy dz + i \int \int \rho_s^* \frac{G^2 \Psi^2 b}{(\bar{u} - a)^2 + b^2} dy dz \right]. \quad (\text{A.7})$$

If the limit of the imaginary part of (6.7) does not vanish dc will be imaginary and unstable solutions will exist. One readily shows that if b approaches zero through positive values then

$$\lim_{b \rightarrow 0^+} \int \int \frac{\rho_s^* G^2 \Psi^2 b}{(u_c)^2 + b^2} dy dz = \pi \oint \rho_s^* G^2 \Psi^2 \left| \frac{\partial \bar{u}}{\partial n} \right|^{-1} dl,$$

where $\partial \bar{u} / \partial n$ is the gradient of $\bar{u}$ normal to the critical curve and $\oint dl$ is a line integral along the closed isopleth $\bar{u} = u_c$. Since the pressure perturbation cannot vanish identically on the critical curve the above integral must be positive, and it follows from (6.7) that $dk^2 < 0$ for $b = \text{Im} dc > 0$. Thus, disturbances which have a wavelength slightly longer than the neutral one are unstable.

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